Quantum Computing: *Implications for Portfolio Managers*

PHIL GODDARD

PHIL GODDARD
is the head of research at 1QB Information Technologies (1QBit) in Vancouver, BC, Canada.
phil.goddard@1qbit.com

Since 1965, when Gordon Moore predicted that the number of transistors on a chip would double approximately every two years, the expectation has been that the physical size of transistors would reduce toward the size of a single atom. As a consequence, researchers began to think about the prospect of developing a computer based on the laws of quantum mechanics.

Over the last 20–30 years, numerous research groups, including ones at International Business Machines (IBM), Microsoft Corporation, and Google, have been working toward the development of a quantum computer. Research labs at major universities are also working toward this goal, with increasing amounts of public and corporate money pouring into the field.

In 2011, D-Wave Systems sold "the world's first quantum computer" to Lockheed Martin. Then, in 2013, they sold an upgraded version to a consortium led by Google and the National Aeronautics and Space Administration (NASA). In 2015, each of those machines was upgraded to the latest "D-Wave 2X" processor. (Undisclosed U.S. government agencies are also believed to be working with the D-Wave processors.)

The majority of the research to date has been focused on the development of hardware required to support the creation and control of quantum computation bits, called qubits. Until recently, little work has focused on how these computers might be used in general and, in particular, how they might be used to solve problems of commercial significance.

The expectation has been that quantum computers will solve problems in orders of magnitude less time than classical computers for a range of applications, including ones in machine learning and cognitive recognition, operations research, and general optimization. This prospect has significant implications in computational finance for solving a broad range of problems, including those found in portfolio analysis, market clique analysis, credit default analysis, and risk management.

This article presents an overview of some of the fundamental concepts of quantum computation, along with its applicability to solving high-value problems in computational finance.

It also presents an overview of a formulation for using existing quantum annealing technology to solve a general portfolio optimization problem. This formulation solves an optimal multiperiod portfolio optimization problem, taking into account transaction costs incurred when rebalancing across time periods. Incorporated into the problem formulation are permanent and temporary market impacts and forecasts of expected future risk and return. This formulation provides, over a multiperiod time horizon, a more optimal portfolio than using a series of single time-step optimal portfolios as is often done today.

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BITS VERSUS QUBITS

The fundamental computational unit of a quantum computer is a quantum bit—or qubit, for short. It is the quantum equivalent of a digital bit in a classical computer.

Whereas a digital bit has a value of either 0 or 1, however, a qubit may have both values at the same time, a property called superposition. Not until the values of the qubits are read, that is, observed, do they settle on a specific value of either 0 or 1. Superposition enables computations performed using qubits to search all possible solutions to a given problem (i.e., all possible combinations of qubits) simultaneously. This process compares with that of a classical computer, which, in order to exhaustively search a solution space, must check all possible combinations of digital bits, often in series. As the problem space increases, the advantage of quantum computing over digital computing becomes more apparent.

Consider a simplified case of finding the minimum value of an optimization problem in which there are 16 possible combinations of binary variables. (One of the 16 combinations corresponds to the optimal solution.) This is called a combinatorial optimization problem and is the type of problem that quantum computers are particularly good at solving.

To encode this problem requires four variables, which may be achieved using either 4 bits (on a classical computer) or 4 qubits (on a quantum computer). Exhibit 1 shows all possible combinations of those bits or qubits. Although the required number of bits/qubits is the same on a classical and on a quantum computer, the number of operations required to find the best solution is very different.

To find the best solution on a classical computer would require all 16 combinations to be checked, typically in series. Alternatively, on a quantum computer, since the qubits are in a superposition of both 0 and 1, all 16 combinations may be checked simultaneously.

EXHIBIT 1
All Possible Combinations of a 4-Variable (16-State) Problem

0000	0001	0010	0011
0100	0101	0110	0111
1000	1001	1010	1011
1100	1101	1110	1111

For a general n-variable problem, there are 2^n possible combinations that must be searched to determine the optimal solution. Even for problems of a size large enough to justify the use of a highly parallelized digital supercomputer, searching all combinations of a general n-variable problem is time consuming. In contrast, all 2^n combinations are evaluated simultaneously on an n-qubit quantum computer in one operation.

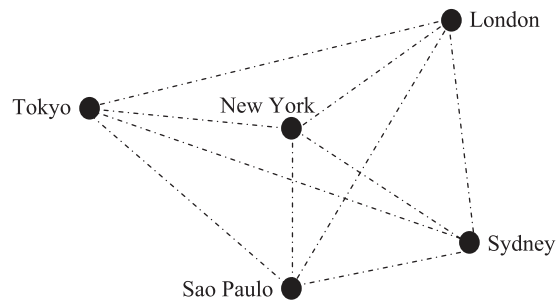
A classic combinatorial optimization problem is the so-called *traveling salesman* problem. Shown in Exhibit 2, the traveling salesman problem is the problem of finding the shortest round-trip path between a set of cities. Finding the optimal travel distance requires evaluating the distance of every possible path between all cities and then selecting the minimum solution. In comparison, since a quantum computer can evaluate all possible path combinations simultaneously, the optimal minimum solution may be determined in one computation.

The traveling salesman problem is very closely related to the optimal multiperiod portfolio construction problem (discussed in the next section). Given a universe of assets and a forecast of expected return and volatility, and taking into account trading costs (including permanent and temporary market impact), all possible combinations of investments must be evaluated to determine the most profitable strategy. Nominally, this can be achieved in one computation using a quantum computer.

The State of Quantum Computing Technology

Broadly speaking, there are two quantum computing paradigms: adiabatic quantum annealing,[1] and gate-model quantum computing—sometimes called universal quantum computing.

EXHIBIT 2
Travelling Salesman Problem



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Conceptually, the gate-model approach is the more general of the two paradigms and the most similar to classical digital computers. An array of qubits, acting like an array of transistors, allows a general computation to be programmed and solved. Owing to the inherent difficulty in designing and controlling an array of gate-model qubits, however, the largest grid thus far demonstrated has been a 4-qubit system (IBM [2015]). The most optimistic estimates of a commercially viable gate-model quantum computer put its availability at 5–10 years in the future.

In comparison, the quantum annealing approach is suitable only for solving specific types of optimization problems. Nonetheless, a 1,152-qubit processor that uses the quantum annealing approach is commercially available today. In 2011, D-Wave Systems supplied its first 128-qubit quantum annealer to Lockheed Martin. Then, in 2013, D-Wave released a 512-qubit processor that was acquired by a joint Google and NASA consortium. Both of those processors were upgraded in 2015 to D-Wave's latest 1,152-qubit annealer. Conceptually, this means that a problem with 2^{1152} (approximately 10^{350}) binary variables can be solved in one calculation.[2]

Initially greeted with skepticism, each new release of a D-Wave processor has resulted in much debate as to whether there are quantum effects taking place in the annealing process and, if there are, whether they contribute to any computational speedup. However, in 2015, Google released results indicating a 10^8 times speedup over classical computations for a very specific type of optimization problem (Denchev et al. [2015]). Results for more general optimization problems are currently being evaluated.

One of the most significant technical issues impeding the development of quantum computers is the presence of *noise*. Taking the form of thermal, electrical, and/or magnetic interference, the more noise that a processor is subjected to, the less likely it is to arrive at a satisfactory solution to any optimization problem. To eliminate thermal noise, the quantum annealer operates at temperatures in the 12–17 milliKelvin range, that is, 12–17 thousandths of a degree above absolute zero—several orders of magnitude colder than interstellar space. The annealer is also housed in a series of enclosures that provides extreme shielding from electrical and magnetic interference.

Working with a classical computer is now so simple that users give little or no thought to the underlying steps involved in taking a general algorithm, typically implemented in a high-level software language or application (such as Python, R, MATLAB, or Excel); solving that algorithm; and displaying the results in a table or graph. Exhibit 3 shows many of the layers involved when working with a classical computer. The ease of use is largely attributable to expertise built up over the last 3–4 decades in writing efficient compilers and understanding how to control and error-correct the state of transistors.

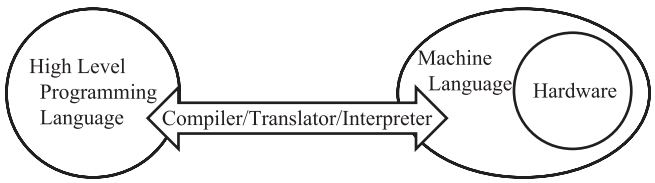
In comparison, the understanding and use of quantum computers is very much in its infancy. To a large extent, using a quantum computer today is similar to using a classical computer 40+ years ago. Specialized expertise is required to write software to translate general optimization problems into a format solvable by a quantum computer. This may mean programming at the machine language level (see Exhibit 3) rather than using a high-level language.

Setting up the problem and tuning the input parameters is a nontrivial and time-consuming task, which requires intimate knowledge of the underlying mathematics of the problem being solved and the characteristics of the quantum annealer.

The 1QBit software development kit (SDK) (1QBit [2016]) provides a set of software tools that allow developers to experiment with using a quantum computer without needing to understand the details of, or the underlying physics of, the annealing processor.

Implemented using C++ and Python, the SDK allows users to access the latest heuristics designed for solving certain types of optimization problems and to build quantum-ready applications, without needing to learn new software languages or become an expert in quantum processor design. In the near future, this SDK is expected to be available to access the quantum annealer through the cloud, allowing optimization problems to be solved remotely.

Exhibit 3
Interfacing with Computational Hardware



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An Overview of Optimization

In general, solving an optimization problem can be thought of as finding a set of parameters that define the minimum of a function, where the function and/or parameters are subject to some constraints. This process is often described in terms of traversing a landscape searching for the lowest point in that landscape.

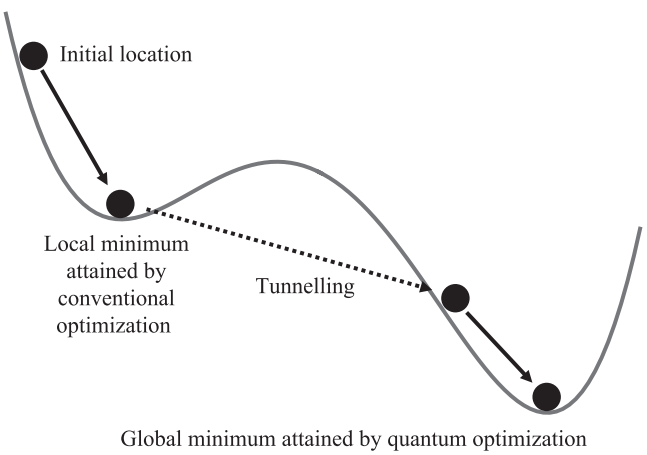
If the landscape is convex—roughly bowl-shaped—then there is a unique global minimum, and any number of algorithms are available to successfully find that minimum. In general, however, the landscape will contain many hills and valleys. This uneven landscape makes finding a global minimum more difficult.

Consider the curve shown in Exhibit 4, where the goal is to find the global minimum of the curve. Assuming an initial starting location in the top left-hand corner, a traditional gradient-based optimization algorithm would search for a minimum by moving down the curve and would find, then *get stuck at*, the local minimum.

For certain types of optimization problems, quantum annealing is believed to have an advantage over traditional optimizers, a notion attributable to a phenomenon called quantum tunnelling. Quantum tunnelling allows an optimizer to more easily search the entire solution space of the optimization problem, thereby having a higher probability of finding the optimal minimum solution.

In the case shown in Exhibit 4, upon reaching the local minimum, quantum tunnelling would allow the solver to tunnel through the landscape and arrive at the global minimum solution, as desired.

EXHIBIT 4
A General Optimization Landscape



The specific type of optimization problem solved by a quantum annealer is called a quadratic unconstrained binary optimization (QUBO) problem. Mathematically, it takes the form

$$\min x^T Q x, \tag{1}$$

where x is a vector of binary variables, and Q is a square matrix. Most traditional optimizers require that Q be an invertible matrix. An additional advantage of quantum annealing is that no assumption on the invertibility of Q is required.

Equation 1 looks very much like the standard Markowitz mean–variance single-period portfolio optimization problem, where x represents the weights of each investment, and Q is a covariance matrix. Yet, there are some important differences. Firstly, the variables are binary (cf. being real valued in the Markowitz case); secondly, there is no assumption on Q being positive definite; and lastly, there can be no constraints placed on the variables.

The process of transforming a generic (real-valued, with constraints) optimization problem into the QUBO form required to solve it using a quantum annealer is not trivial. It requires a good understanding of how to effectively embed the problem into the annealing chip and then tune the parameters to achieve the best performance. This transformation is one of the components that is built into 1QBit's SDK; it removes the need for users to understand the complexities of the embedding task while retaining the benefit of using the quantum annealer.

AN OPTIMAL MULTIPERIOD TRADING TRAJECTORY

Given the large number of optimization problems encountered in computational finance, it is reasonable to expect that finance professionals will be among the earliest adopters of the technology.

In this section, an overview of an optimal multiperiod portfolio optimization problem that has been solved on a quantum annealer is presented. (For a detailed mathematical derivation and implementation details, see Rosenberg et al. [2016].) A comparison is made with traditional Markowitz mean-variance portfolio optimization;

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a description of the conditions under which circumstances a discrete variable solution is preferred is given; and a method for creating such a discretized solution is presented.

Consider an asset manager who wishes to invest K dollars in a universe of N assets with an investment horizon divided into T time steps. Exhibit 5 shows an idealized solution to this problem in a four-asset universe over a time horizon of eight steps, where ten units must be invested at any given time.

Given a forecast of expected future returns and the risk of each asset at each time step, an asset manager must decide how much to invest in each asset, at each time step, while accounting for transaction costs. This process forms a trajectory through time that the asset manager should follow to maximize portfolio returns. In general, the transaction costs should include not only the up-front immediate cost of buying or selling an asset, but also any permanent or temporary impacts that that transaction may have on market prices.

A naïve approach to this problem is to solve for the mean–variance portfolio that maximizes the expected return subject to a level of risk at each time step. The resulting portfolios form a set in which each portfolio is locally optimal at a given point in time.

Still, this approach fails to account for rebalancing costs. This means that it is highly likely that there will be a different set (i.e., a different trajectory) of portfolios that will be globally optimal in the sense that the overall return achieved by following the given trajectory will be greater than the return from following the series of single time-step optimal portfolios.

This multiperiod portfolio trajectory problem can be thought of as a combinatorial optimization problem of a form similar to the traveling salesman problem discussed earlier (see Exhibit 2). The optimal trajectory may be found by calculating the return of all possible trajectories and selecting the one that provides the greatest return. Hence, after some suitable modifications, it is amenable to being solved using a quantum annealer.

The weights derived from a traditional mean–variance portfolio optimization problem are continuous, that is, the weights will take on decimal values. For assets that can only be traded in large lots, or for asset managers who are constrained to trading large discrete blocks of assets, however, solving the continuous-weight problem, then rounding to the nearest integer block size (as is often done today), leads to a suboptimal approximation. In this case, solving a discrete version of the problem, that is, in which the weights are constrained to have only integer values, is likely to lead to a better trajectory of portfolios.

Examples in which a discrete solution are required are (a) when an institutional investor is limited to trading even lots owing to a prohibitive premium on trading odd lots, or (b) when an asset manager invests in baskets that trade only in integer lots such as ETF-creation and ETF-redemption baskets, which trade only in large multiples, such as fund units of 100,000 shares each.

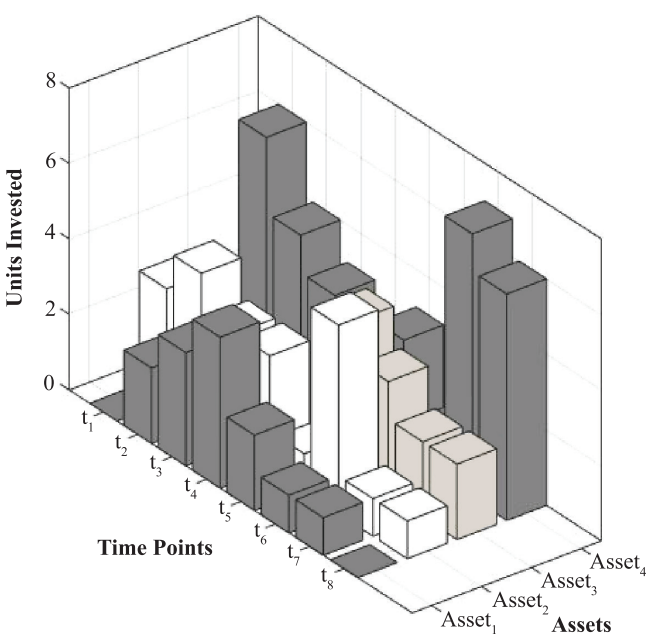
The discrete problem is nonconvex because of the fragmented nature of the solution domain and is therefore much harder to solve than the traditional continuous weight problem. Hence, there are advantages to using a quantum annealer for determining an optimal solution.

Another advantage of using a quantum annealer is that, in general, the covariance matrix describing the relationship between the universe of assets may be ill-conditioned, or degenerate. This condition would typically cause traditional optimizers to fail. However, ill-conditioning, or degeneracy, does not pose a problem when solving the optimization problem using the quantum annealer.

Before solving optimal trading trajectory problems using the quantum annealer, the variables must be converted from the discrete integer formulation of

Exhibit 5

A Discrete Multiperiod Trading Strategy



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the problem statement to the QUBO problem's binary formulation (Equation 1). There are several ways that this encoding can be performed, each having a different impact on the size of the problem (i.e., the values of K, N, and T) that can be solved. A significant consequence of this transformation is that it imposes a constraint on the size of problem that can currently be solved.

Numerical Results

As discussed previously, unlike on a classical computer, to obtain a good solution using a quantum annealer requires a good understanding of how to transform a general problem into the format required by the hardware. There are multiple hardware settings that can be tuned—all of which affect the quality of the result—and multiple ways of embedding the problem into the chip (i.e., utilizing the qubits in the most efficient way) that affect the size of the problem that can be solved. See Rosenberg et al. [2016] for a detailed discussion of the effect of these settings and a range of results.

Exhibit 6 shows a small but representative sample of the results from Rosenberg et al. [2016]. Here, N is the number of assets in the universe, T is the number of time periods under consideration, K is the number of units that must be invested at each time period, and $S(\alpha)$ is the success rate, where success is measured as being within α% of the optimal solution.

It is important to note with these solutions that the quantum annealer is a heuristic solver. That is, it is not guaranteed to give the optimal solution and may give a (slightly) different answer each time the same problem is solved. Hence, each problem is solved multiple times (in this case 200 times) and the solution that is attained the most frequently is taken as the final solution.

As shown in Exhibit 6, existing quantum annealing technology allows only small problems—that is, a small number of assets, time periods, and units to be invested—to be solved. Yet, it demonstrates that existing quantum annealing technology can lead to answers that are equal to, or very close to, the optimal solution.[3]

EXHIBIT 6
Accuracy of Calculated Portfolios

N	T	K	$S(0)$	$S(1)$	$S(2)$
2	3	3	1	1	1
3	2	3	0.99	1	1
2	6	3	0.8	1	1
3	6	3	0.15	0.58	0.92

CONCLUSIONS

Computers that leverage the power (and mysteries) of quantum mechanics are available today, and for some specialized problems, they have been shown to provide significant speed advantages over classical computers.

Industry leaders and researchers are starting to investigate the use of these computers for solving commercially valuable problems using techniques from the fields of machine learning, operations research, and optimization. Through this work, they are laying the groundwork for solving high-value problems in computational finance, including portfolio and risk management.

This article has presented an overview of some of the key concepts and opportunities in quantum computing, and applied them to solve a multiperiod portfolio optimization problem. Although the size of the portfolio optimization problem currently solvable is small (in terms of the number of investable assets, time horizon, and investable units that can be handled), the formulation presented holds for very generic and large-scale problems. Importantly, it allows temporary and/or permanent market impacts to be incorporated in the optimization when large blocks of assets must be traded.

Today, it is possible to write algorithms that can be solved using state-of-the-art classical computers, but are ready—by the simple flick of a (software) switch—to be solved by a quantum computer. As quantum computing technology improves in the near future, it is expected that it will revolutionize the way that optimization is performed in computational finance.

ENDNOTES

[1]*Annealing* is named after the process of heating and subsequent slow cooling of a material, typically a metal, in an attempt to preserve certain properties, such as strength, by reducing internal stresses on the atoms that make up the material.

[2]By comparison, there are approximately *only* 10^{80} atoms believed to exist in the known universe.

[3]For problems of this size, the optimal solution may be calculated on a classical computer using an exhaustive search of all possible trajectories. For testing purposes, the solution obtained by the quantum annealer can then be compared with the known optimal solution.

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